Self-Consistent 2-D Model for Quantum Effects in n-MOS Transistors

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***Abstract*— We present a self-consistent two-dimensional (2-D) model for carrier quantization effects in the channel of highly-doped n-MOSFET's. Quantization is taken into account inside a box region surrounding the inversion channel. The proposed approach extends previously proposed one-dimensional (1-D) schemes allowing to estimate the quantum mechanical (QM) effects on the device current. Good convergence properties are achieved exploiting the effective intrinsic density concept. The simulator has been applied to MOS devices with different peak channel doping, resulting in an improved description of the device behavior.**

***Index Terms*— MOS devices, quantization, semiconductor device modeling.**

I. Introduction

THE push toward high integration densities, low power dissipation, and high speed has led the silicon electronics industry to a continuous scaling of the MOS technology. As the channel length is reduced, however, a number of problems arise, generally known as short-channel effects. To curb these effects, and at the same time to increase the current drive, a thin gate oxide and a high substrate doping concentration near the silicon-oxide interface must be employed. This, in turn, implies a high transverse electric field at the interface, even at the onset of the inversion. The strong electric field in the inversion layer gives rise to a potential well which can become sufficiently narrow to cause significant quantization effects in the motion of carriers perpendicular to the interface. In these conditions the conduction band can be no longer regarded as a continuum of states, but rather splits into discrete subbands. The calculated quantum mechanical (QM) charge density peaks a few angstrom away from the interface, resulting in a reduction in the gate capacitance, which can be equivalently seen as an increase in the effective oxide thickness. Moreover, the transport properties of the confined electrons, e.g., their mobility, differ from those valid in the bulk. These effects can influence the device electrical characteristics, in particular the threshold voltage and terminal currents, and must be accounted for to maintain a reliable description of the device behavior. Conversely, parameters like the oxide thickness and polysilicon active doping extracted from experimental data without considering QM effects can be affected by significant errors.

The calculation of the electronic properties of the two-dimensional electron gas (2-DEG) and their effects on the device characteristics has been previously carried out in a one-dimensional (1-D) approximation by several groups [1]–[8]. However, because of the 1-D approach, no direct information on the current flow could be obtained, and the threshold voltage shift had to be inferred from 1-D capacitance–voltage (C–V) simulations. A 2-D implementation based on a simplified quantization model has been reported [9], but results are shown only for the gate capacitance. In this work we report a 2-D extension of the 1-D model and its implementation in a general-purpose device simulator. In the next section we describe the 1-D numerical procedure, while in Section III the extension to 2-D simulations is presented. The convergence properties of the proposed algorithm are investigated. Results of the simulation of highly-doped MOSFET's in the linear and saturation region are presented in Section IV.

II. One-Dimensional Model for Carrier Quantization

The study of the 2-D behavior of the carriers in the inversion layer of MOS transistors raises interesting problems in many fields: from the electrical characterization to the physical analysis to the numerical modeling. The generally adopted assumption is the homogeneity of the potential well in the plane parallel to the interface. Therefore, each wave function can be expressed as a product of a 2-D plane wave parallel to the interface (xy plane) and an envelope function determining the confinement of the electrons in the direction normal to the interface (z direction). The calculations are then performed within a standard effective-mass approximation in a parabolic band, leading to the equation for the envelope function of the jth level in the ith valley, ζ_{ij}

$$-\frac{\hbar^2}{2}\frac{d}{dz}\left(\frac{1}{m^*_{i,z}}\frac{d\zeta_{ij}}{dz}\right)+E_p\zeta_{ij}=E_{ij}\zeta_{ij} \tag{1}$$

where E_p is the potential energy of the electrons, E_{ij} and ζ_{ij} are the eigenenergy and the associated eigenfunction of the ith valley of the jth subband, and $m^*_{i,z}$ is the effective density of states mass in the ith valley along the direction perpendicular to the interface. Once the energy levels have been determined, the electron density must be computed. Assuming Fermi–Dirac statistics, $f(E)$, we have

$$n(z)=\sum_{ij}|\zeta_{ij}|^2\int_{E_{ij}}^{\infty}f(E)D_{ij}(E)\,dE \tag{2}$$

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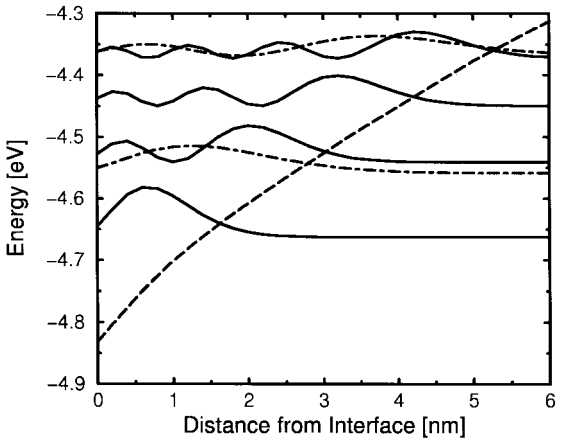


Fig. 1. View of the lowest six energy levels and corresponding squared envelope functions for a device with peak channel doping of 2.5×10^{18} cm^{-3} and a gate oxide thickness of 100 Å. The gate bias is 5 V. The dashed line is the potential well energy, solid lines are used for the longitudinal levels and dot-dashed lines for the transverse levels.

where $D_{\rm ij}$ is the 2-D density of states for electrons, given by

$$D_{\rm ij}(E) = \frac{g_i m^*_{i,xy}}{\pi\hbar^2} H(E - E_{\rm ij}). \tag{3}$$

In the last equation, $H(E)$ is the Heaviside function and g_i is the degeneracy factor of the ith valley. For (100) silicon, we have two longitudinal valleys with $m^*_{i,xy} = m_t$ and four transverse valleys with $m^*_{i,xy} = \sqrt{m_l m_t}$ [5]. Inside the oxide, we have instead taken $m^* = 0.5 m_0$ [10].

Equation (1) cannot be solved analytically for a generic potential energy profile. We used a matrix eigenvalue solution routine which provides at the same time both the eigenenergies and the associated eigenfunctions. A typical potential well profile is shown in Fig. 1, where the squared envelope functions $|\zeta_{\rm ij}|^2$ corresponding to the six lowest energy levels are pictorially shown, each referred to the respective level.

For the typical range of doping and bias conditions, (1) can be solved in a region extending from a few angstrom inside the oxide to a few nanometers inside the silicon. At both ends, the appropriate boundary condition for (1) is to assume a vanishing wave function. It can be expected that the resulting eigenenergies will not be influenced by the selected region depth as long as their levels lie sufficiently below the well energy value at the substrate end of the solution domain. This condition may not hold for the higher energy levels, particularly for the case of low transverse electric field. In turn, this would lead to inaccuracies in the computed charge density, since a significant fraction of the electron charge is contributed by these levels. A solution to this problem could be to select a variable thickness of the quantum solution domain, e.g., limited by the depleted region itself [11]. However, this could result in an unnecessarily large solution region, and ultimately in a waste of computational time. Therefore, we followed an alternative approach as described in [12]: we retained a fixed number of subbands, namely three, and employed a classical approximation for the energies above the fourth level. In this framework, the quantum-mechanical electron density, n_q, is given by

$$n_q(z) = \frac{kT}{\pi\hbar^2} \sum_{\rm ij} g_i m^*_{i,xy} \cdot \log\left[\frac{1+\exp\left(\frac{E_{\rm fn}-E_{\rm ij}}{kT}\right)}{1+\exp\left(\frac{E_{\rm fn}-E_{cl}}{kT}\right)}\right] |\zeta_{\rm ij}|^2 \tag{4}$$

where $E_{\rm fn}$ is the electron quasi-Fermi level and E_{cl} represents the energy above which the electron density is described classically. This corresponds to the lower energy between that of the fourth level and that of the highest bounded level. The classical "tail" of the electron distribution, n_t, is instead described by the usual Maxwell–Boltzmann statistics

$$n_t(z) = N_c \exp\left(\frac{E_{\rm fn}-E_{cl}}{kT}\right) \tag{5}$$

where N_c is the density of states in the conduction band. This approach allows to maintain an accurate description of the overall charge while keeping the quantum region within reasonable limits. For the results shown, the solution domain is 40 nm, with a mesh spacing close to the interface of 0.2 nm. With this conservative choice (and with the very strict convergence criterion adopted, see Section III) we could obtain a very accurate numerical solution, at the cost of a relatively high computational time: typically an I_D–V_G characteristic with 50 bias points runs overnight for a device with a few thousand meshpoints on a medium-range workstation. However, we have verified that reducing the depth of the domain for the quantum solution does not result in a significant modification of the overall charge density; therefore, a significant improvement in CPU performance can be obtained whenever a sub-millivolt accuracy is not required.

In Fig. 2 the two contributions to the channel electron profile are shown for a device with peak channel doping of 2.5×10^{18} cm^{-3}, a gate oxide thickness of 100 Å and a gate bias of 5 V. Note that the classical part of the charge is clipped where the energy level E_{cl} crosses the conduction band (Fig. 1), and that this approach naturally provides a smooth transition between the QM and classical distributions toward the bulk. In Fig. 3 the same contributions are shown for a typical charge distribution inside the drain diffusion. Note that in this region the electrons pertaining to the classical tail represent a much larger fraction of the overall charge than inside the channel, i.e., the carriers quantization is much weaker.

III. The Two-Dimensional Model

The physical insight that can be gained from a simple 1-D model is limited when application to real devices is concerned. In these cases, to obtain reliable estimates of quantities like drain current, transconductance, and gate capacitance, the adoption of at least a 2-D model is required. In this context, several issues must be addressed, among which the convergence of the numerical algorithm and the transition between the channel and the source and drain regions, where QM effects vanish and the classical solution must be recovered.

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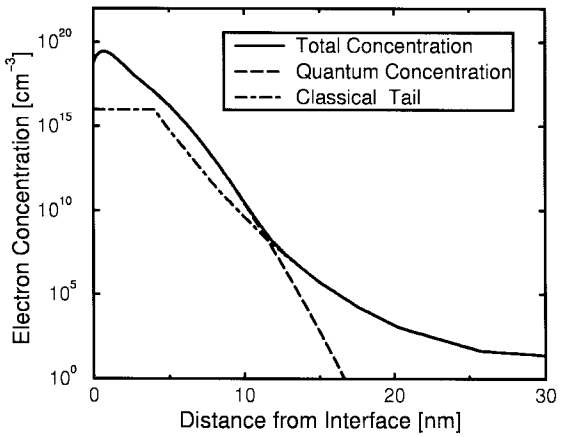


Fig. 2. Channel electron concentration for the same device and bias conditions as in Fig. 1. Note the two contributions to the QM charge.

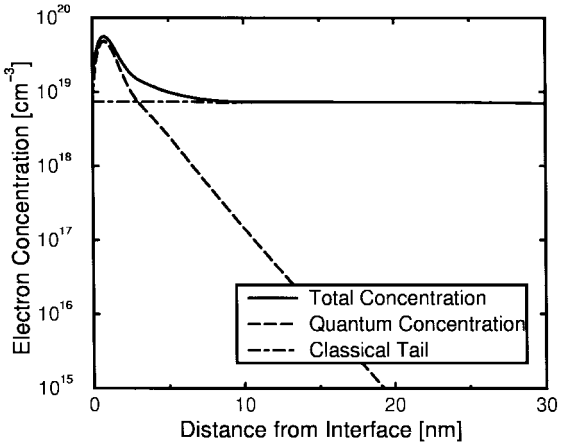


Fig. 3. Electron concentration as obtained from a potential energy profile typical of the drain region in an MOS transistor.

Our approach was to take into account QM effects only in a rectangular region including the inversion channel, and also a part of the source and drain diffusions. In other words, the box region is made wider than the metallurgical channel length (see Fig. 4). This allows to actually solve the Schrödinger equation only in the device regions where the effect is expected to be important.

From a physical point of view, it is expected that as long as the potential energy variations along the channel are much smaller than in the z direction, the Hamiltonian can be decoupled into a z-dependent term describing the quantization effects in the z direction, and a term representing the plane wave motion along the interface plane, where no quantization occurs. This approximation provides the base for all the 1-D treatments available in the literature, and actually holds until the potential energy variation along a few carrier scattering lengths (i.e., about 20 nm) becomes comparable with the confining potential well [13]. Hence, even for deep-submicron devices biased under high drain-source voltage conditions, it is justified to describe the 2-D solution of the Schrödinger equation by interpolating a set of 1-D solutions in direction parallel to the interface plane. A schematical view of the sections is also reported in Fig. 4.

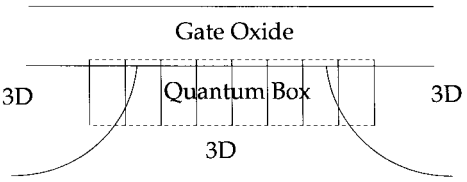


Fig. 4. Simplified geometry of a typical MOS device. The box region where the QM behavior of electrons is explicitly described is shown.

Consistency between the Schrödinger and Poisson equations has proven to be a challenging numerical problem when the computed QM charge distribution is directly inserted into the Poisson solver. When a 2-D simulation is concerned, moreover, additional problems may arise because of the carrier continuity equations. In fact, the state variable associated with both the electron continuity equation and the QM charge distribution model (1)–(3) is the electron density. Therefore, a severe numerical conflict would occur, due to the implicit ill-posedness of the problem. A way to circumvent this problem is to note that conventional device simulators build up the charge using three-dimensional (3-D) Maxwell–Boltzmann statistics, in a way similar to (5). Furthermore, the intrinsic concentration n_i is classically related to the bandgap value. Since quantization effects introduce a modification in the structure of the (classical viewed) conduction band, raising the lowest occupied level and thus widening the effective bandgap value, we can describe this phenomenon with a local modification of the intrinsic carrier density [12], [14]. We have therefore defined an *effective* intrinsic density, n_i^*, related to the ratio between the quantum charge (i.e., $n_q + n_t$) and the classically computed electron charge density

$$n_i^* = n_i \frac{n_q + n_t}{n_{cl}}. \tag{6}$$

It can be easily seen that a standard simulator based on 3-D occupation statistics and relying on this value for the intrinsic concentration obtains the correct quantum-mechanical charge profile for the same electrostatic and quasi-Fermi potentials. Note that in a 2-D simulation this algorithm is, to our best knowledge, the only viable way to incorporate QM effects on the current flow within a classical framework. Of course this approach can also be applied to 1-D simulations; in particular, in the channel of a MOSFET, where the current is assumed to flow perpendicularly to the considered section (i.e., the quasi-Fermi levels are taken as constant), it is no longer necessary to solve for the continuity equations. As a consequence, the ill-posedness is removed. Nevertheless, it has been demonstrated that our approach grants better convergence properties [14] when compared with more conventional methods where the QM charge is directly inserted into the Poisson equation [15], [16]. Note moreover that the proposed approach is similar to that used for implementing Fermi–Dirac statistics in Maxwell–Boltzmann based device simulators [17]. Another interesting feature of this approach lies in its extreme simplicity, in that it does not require any modification of the numerical algorithms of the simulator, but just of its input/output routines. Following this guideline, for each grid

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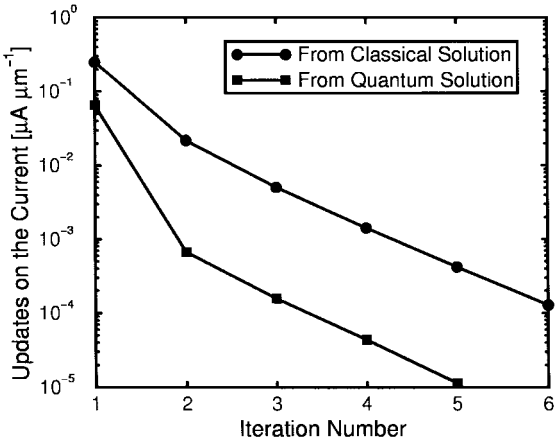


Fig. 5. Updates on the drain current of an n-MOS biased at $V_D = 0.1$ V, $V_G = 5$ V when the initial guess is a classical solution computed for the same bias point and a converged QM solution for the same drain bias and $V_G = 4.9$ V.

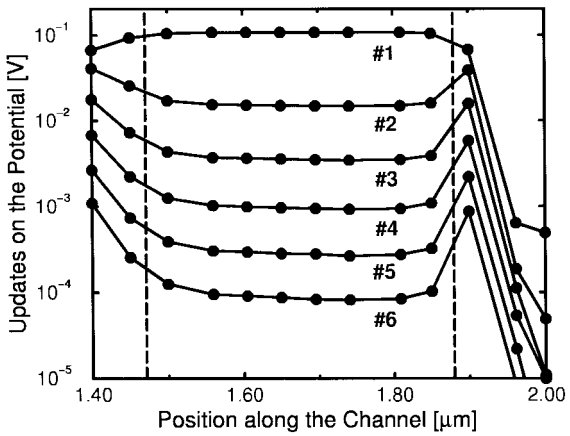


Fig. 6. Maximum value of the updates on the electrostatic potential for the different sections along the channel. Bias conditions are $V_D = 0.1$ V, $V_G = 5$ V. A classical solution was taken as initial guess. The dashed lines identify the source and drain metallurgical junctions.

node inside the selected box region, a correcting factor for the intrinsic carrier density is computed as

$$\gamma = n_i^*/n_i. \tag{7}$$

This factor is used as an input to the device simulator to compute a new intrinsic carrier density and, in turn, an improved solution. The procedure is iteratively repeated until convergence is reached.

In Fig. 5 the convergence rate of this algorithm is shown for two different initial guesses: a classical solution at the same bias and a QM solution computed at a previous bias point. This latter case if of importance when an entire device characteristics must be generated. Note that, although the achieved convergence rate is exponential in both cases, in the latter—as expected—the update on the drain current requires a smaller number of iterations to reach a fixed value. A deeper insight into the convergence properties can be gained by looking at the behavior of the electrostatic potential. With respect to a global indicator like the drain current, which is

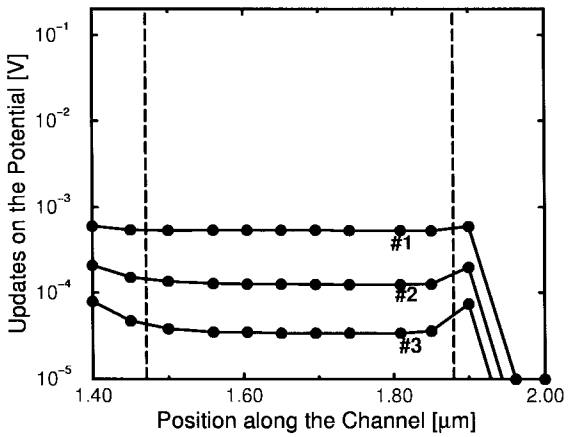


Fig. 7. Same as in Fig. 6 but with a previously computed QM solution at a smaller bias taken as initial guess.

due to conduction in the whole inversion channel, the updates of the potential $\Delta\psi(x, z)$ give a detailed information on the location of the regions which limit the convergence. In Fig. 6 we show the updates on the electrostatic potential as a function of position x along the channel, for the first six iterations and calculated assuming a classical initial guess. For each x-coordinate, the maximum value of the updates along z is shown, which almost invariably corresponds to the silicon/oxide interface node. Note that inside the inversion channel the error drops rapidly, the convergence being limited by the sections located inside the source and drain diffusions. In these regions, in fact, a transition to a classical solution occurs, and the effect of the nonuniformities of the charge distribution is noticeable. The updates on the potential are much more uniform if a QM solution at a lower bias is taken as the initial guess (Fig. 7), since in this case the QM behavior of the charge is already established in the critical source and drain regions. In our simulations, the convergence criterion was $\|\Delta\psi\|_\infty < 100$ μV. With this very strict requirement and high gate bias (5 V), it can be seen from Fig. 6 that convergence from a classical solution is reached in about ten iterations. Instead, three iterations usually suffice when a QM solution computed for a gate voltage 0.1 V smaller is used as initial guess. However, if only the terminal current values are of interest, taking advantage of the rapid and monotonical decrease of the error shown in Fig. 5, a small, fixed number of iterations can be used. To further investigate this issue, we computed a complete I_D–V_G transfer characteristic at a drain bias of 100 mV with one and two iterations. In Fig. 8 the relative error on the drain current referred to the converged QM value is shown. The maximum error in the two cases is respectively 14% (at $V_G = 2.6$ V) and 0.046% (at $V_G = 1.6$ V). This is to be compared with the error that would be obtained by retaining the classical solution, that is 2600%. However, the relative error is not very meaningful in the subthreshold region; if only the current behavior above threshold ($V_T = 3.2$ V, marked by an arrow in Fig. 8) is considered, the maximum errors are 4% and 0.02% with one and two iterations, while the error for the classical solution is larger than 10% for all bias points. With three iterations the maximum relative error (not shown in Fig. 8) further

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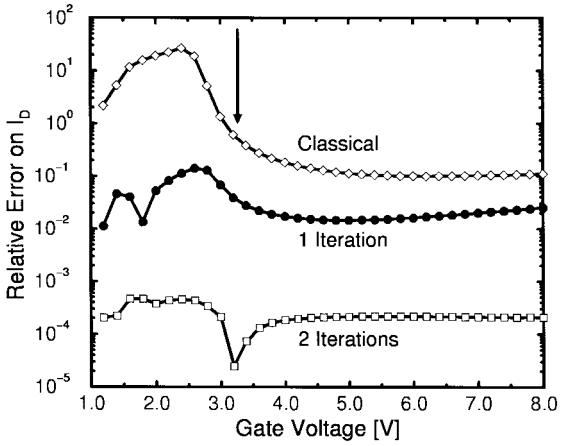


Fig. 8. Relative errors on the drain current for the classical solution and QM solutions with a fixed number of iterations. The ramps are calculated with a 200 mV step. The arrow marks the threshold voltage.

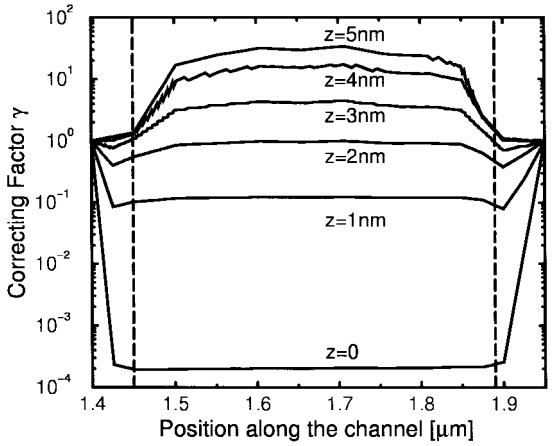


Fig. 10. Correcting factor γ as a function of the position along the channel, taken at different distances from the silicon/oxide interface.

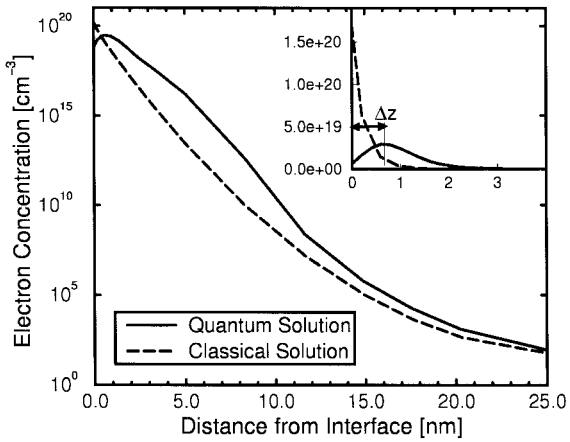


Fig. 9. Charge distributions according to a classical solution and a quantum mechanical solution, after convergence is reached. The inset shows the same data on a linear scale.

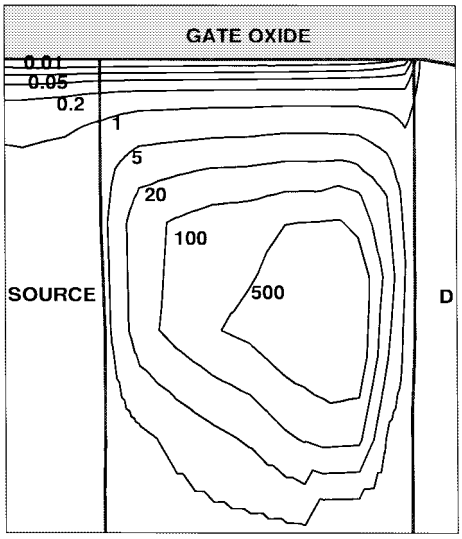


Fig. 11. Two-dimensional view of the correcting factor for the intrinsic density. The vertical axis has been expanded by a factor 50; the metallurgical channel length is 0.45 μm, and the portion of the channel shown is 140 Å deep. Bias conditions are $V_D = 5$ V, $V_G = 5$V.

reduces to less than 0.01%. This clearly demonstrates that two iterations suffice if only the drain current is to be computed.

IV. MOSFET Simulation

To verify the impact of channel quantization on realistic devices, we simulated four n-MOSFET's with peak channel doping ranging from 4.5×10^{17} to 2.5×10^{18} cm^{-3}, an oxide thickness of 100 Å and a metallurgical channel length of 0.45 μm.

A. Charge Distribution

The results presented in this subsection refer to the device with the highest doping; unless otherwise noted, the drain bias is 0.1 V. In Fig. 9 the classical and the quantum-mechanical charge distributions are compared for a section inside the channel region. Close to the interface, the electron distribution has the typical QM profile already shown in Fig. 2. However, due to the rearrangement of the conduction band energy to accommodate for the displaced peak in the QM charge, the electron density is modified even farther away from the interface. In Fig. 10 the resulting intrinsic density correcting factors γ taken at different distances z from the interface are shown as a function of the position x along the channel, with the device biased in the linear region. As can be seen from Fig. 9, the largest relative difference between the QM and classical distributions is located at the interface, where the QM charge almost vanishes. Therefore, in this region, the n_i ratio, i.e., γ, drops by orders of magnitude. Since in this case the voltage applied between source and drain is low ($V_{DS} = 0.1$ V), the correcting factor exhibits a nearly constant behavior along the channel. Moving inside the source and drain diffusions, the energy quantization effect tends to vanish and the intrinsic density relaxes to the classical value. The same information is shown in Fig. 11 as a 2-D contour plot of the correcting factor γ. Inside the source/drain diffusions, a significant modification of the intrinsic density takes place very close to the interface, where a weak quantization effect

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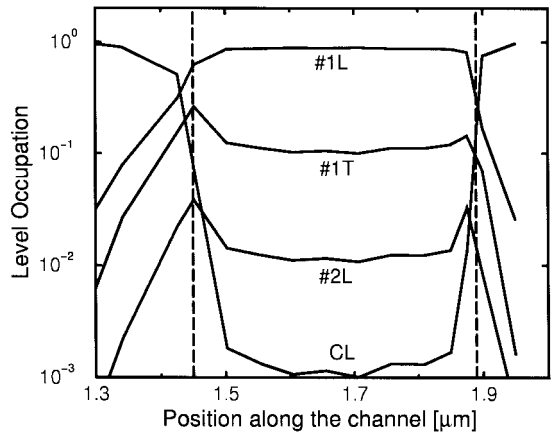


Fig. 12. Charge sharing among the quantum energy levels and the classical tail, as a function of the position along the channel. Labels denote the level number and the valley orientation (longitudinal or transverse). CL denotes the classical tail.

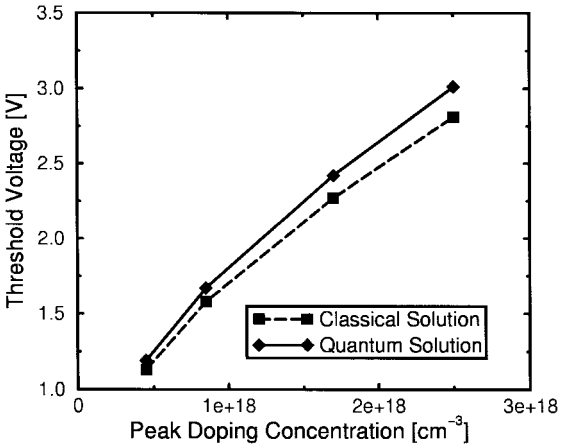


Fig. 13. Simulation results for the threshold voltage as a function of peak channel doping, with the classical and QM model. The increasing threshold voltage shift can be observed.

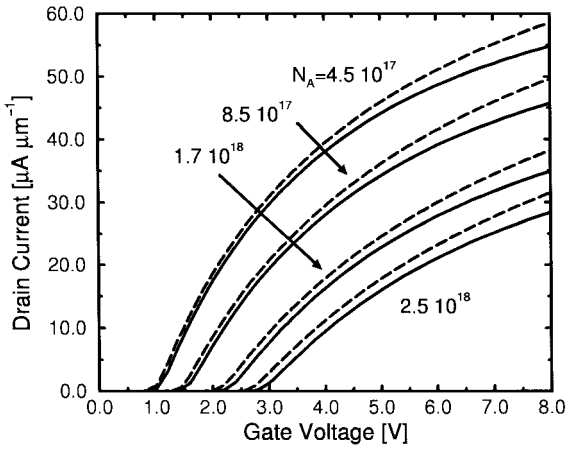


Fig. 14. Drain current as obtained from classical simulations (dotted lines) and QM simulations (solid lines). The drain bias is 100 mV.

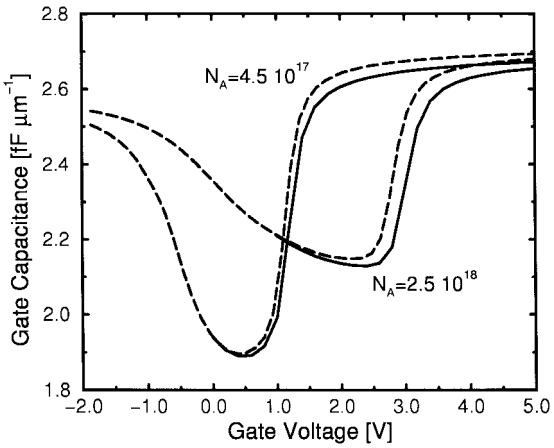


Fig. 15. Comparison between classical and QM-computed gate capacitance for two devices among those shown in Fig. 14.

is still present. In Fig. 12 the percentage occupation of the charge among the different levels in strong inversion condition is shown. Inside the inversion layer more than 99% of the charge is contained in the first three subbands, so that the classical tail approximation of (5) does not significantly affect carrier quantization.

B. Model Results

The threshold voltages obtained from a purely classical and a QM simulation of the above-mentioned devices are shown in Fig. 13; as expected [12], the threshold voltage shift increases with channel doping. In Fig. 14 the drain currents obtained from the classical and QM simulations are shown, for a drain bias of 100 mV. Due to the dependence of the electron mobility on the normal electric field, the QM current is not simply shifted, but also significantly distorted because of the different self-consistent electrostatic potential distribution. The total gate capacitance for the same bias conditions is shown in Fig. 15 for the devices with the lowest and highest doping. The threshold voltage shift and the gate capacitance reduction in strong inversion, due to the increased distance of the mobile charge from the interface, can be clearly observed. Note moreover that the QM-computed gate capacitance smoothly recovers the classical value in the depletion region, where quantization effects are indeed negligible. Since for these devices the geometry is strongly 2-D, the application of a 1-D approximation [12], [14] cannot reproduce both the strong inversion limit and the depletion bias conditions (i.e., the minimum value). Results for the high-drain bias condition ($V_D = 2$ V) are shown in Fig. 16. Note that the proposed algorithm allows to reach numerical convergence also for a saturated channel. A more detailed comparison of the experimental data with the simulation results will be shown in a forthcoming paper [18], where the implementation of a suitable low-field mobility model taking into account channel quantization will be presented. Finally, we checked the usefulness of our 2-D QM approach even on relatively long (0.45 μm metallurgical channel length) devices by comparing the corresponding drain current with the classical one shifted and scaled for the decrease in the gate capacitance, where both corrections have been calculated with a self-consistent 1-D QM simulation. The maximum relative error achieved with this hybrid approach is about 2% in the linear region ($V_D = 0.1$ V) and 5% in the saturated region ($V_D = 2$ V),

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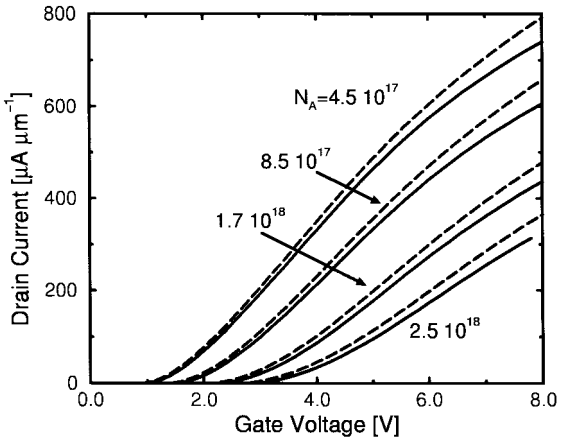


Fig. 16. Drain current for the same devices as in Fig. 14 as obtained from classical simulations (dotted lines) and QM simulations (solid lines). The drain bias is 2 V.

to be compared against a relative error of about 10% (see Fig. 8) for the purely classical result. As expected the QM current cannot be simply computed by a post-processing of the classical curve, even when the threshold voltage shift and the change in the gate-body capacitance are known. In fact, this would neglect the intrinsically 2-D dependence of the total gate capacitance and of the electron mobility on the self-consistent electrostatic potential, which can be correctly described only by a full 2-D QM simulation.

V. Conclusions

A 2-D self-consistent simulator for QM effects in n-MOSFET devices has been presented. The quantum behavior of electrons is explicitly considered in a rectangular region extending from the gate oxide interface to 40 nm inside the silicon and penetrating slightly into the source and drain diffusions. The 1-D Schrödinger equation is solved in the direction perpendicular to the interface along a few sections, and a local correcting factor for the intrinsic charge density, γ, is used to describe the QM behavior; then an interpolation in the direction parallel to the silicon/oxide interface is performed. Only a fixed number of levels are populated according to the 2-D density of states, while a 3-D statistics is used for the upper energies. The iterative algorithm converges very quickly: even just one iteration provides a reasonable estimate of the QM effects (i.e., the threshold voltage shift), while with two iterations very accurate results for the I–V characteristic can be obtained. Results for the threshold voltage and the drain current, both in linear and in saturation conditions, are shown for 0.45-μm MOSFET's with different channel doping levels. The model allows for a physically-based evaluation of QM effects and can therefore be a valuable design tool for the channel profile and junction architecture engineering of highly-doped MOS devices.

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